Graphs & Diagrams

Engineering Chemistry (UE26CY151A) · Semester 1 · Unit 1 · Compiled from PESU AV-summary transcripts, class slides, the Unit-1 slide PDF and past ESA papers · 13 Oct 2026

★★★ high · ★★ medium · ★ low/conceptual weight | Not on current slides Transcript only Added for clarity | source tags: [S004·8] slide, [T005] transcript, [PDF] 137-page deck, [ESA] past paper

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M1 · Molecular Spectroscopy

T1 · EM radiation & interaction with matter

No graph for this topic (see formulas/notes).

T2 · Electromagnetic spectrum & regions

No graph for this topic (see formulas/notes).

T3 · Born–Oppenheimer approximation

No graph for this topic (see formulas/notes).

T3b · Beer–Lambert law

No graph for this topic (see formulas/notes).

T4 · Rotational (microwave) spectroscopy

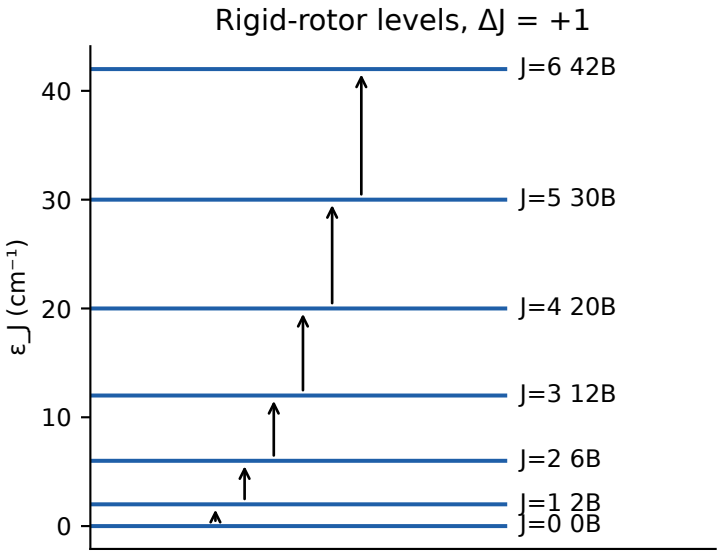


Fig. G9 — Rigid-rotor energy levels J=0–6 (values J(J+1)B), ΔJ=+1 arrows.

In an exam: Label J, the energy in B, and the widening gaps 2B,4B,6B…

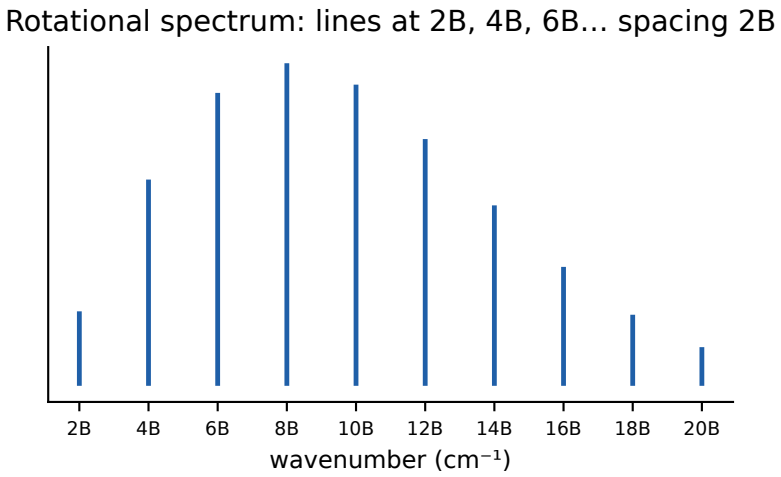


Fig. G10 — Rotational stick spectrum.

In an exam: Lines at 2B, 4B, 6B…; constant spacing 2B; intensity rises then falls.

T5 · Vibrational (IR) – SHO

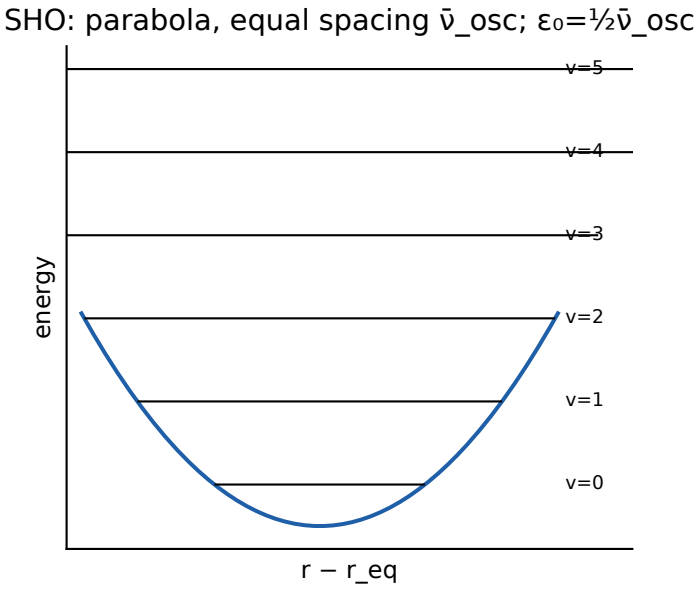


Fig. G13 — SHO parabola with equally spaced levels.

In an exam: Mark v=0…5 and $\varepsilon_0=\frac{1}{2}\bar{\nu}$; spectrum is a single line.

T6 · Vibrational – anharmonic

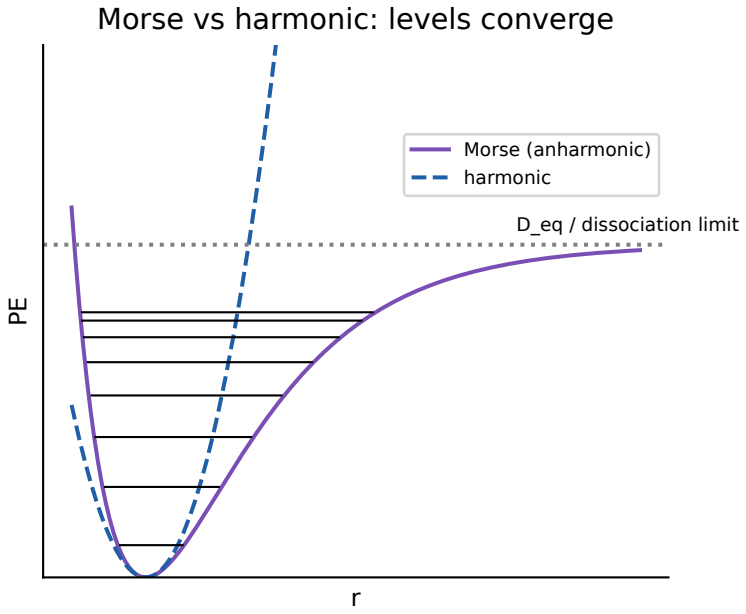


Fig. G15 — Morse curve vs harmonic parabola.

In an exam: Show D_eq, convergence of levels and dissociation limit.

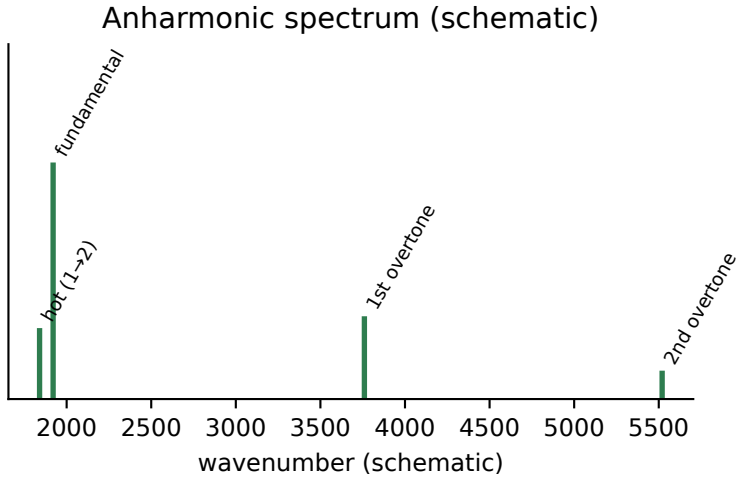


Fig. G17 — Anharmonic IR spectrum (schematic).

In an exam: Hot band left of fundamental; overtones with decreasing spacing.

T7 · Electronic spectroscopy & Franck–Condon

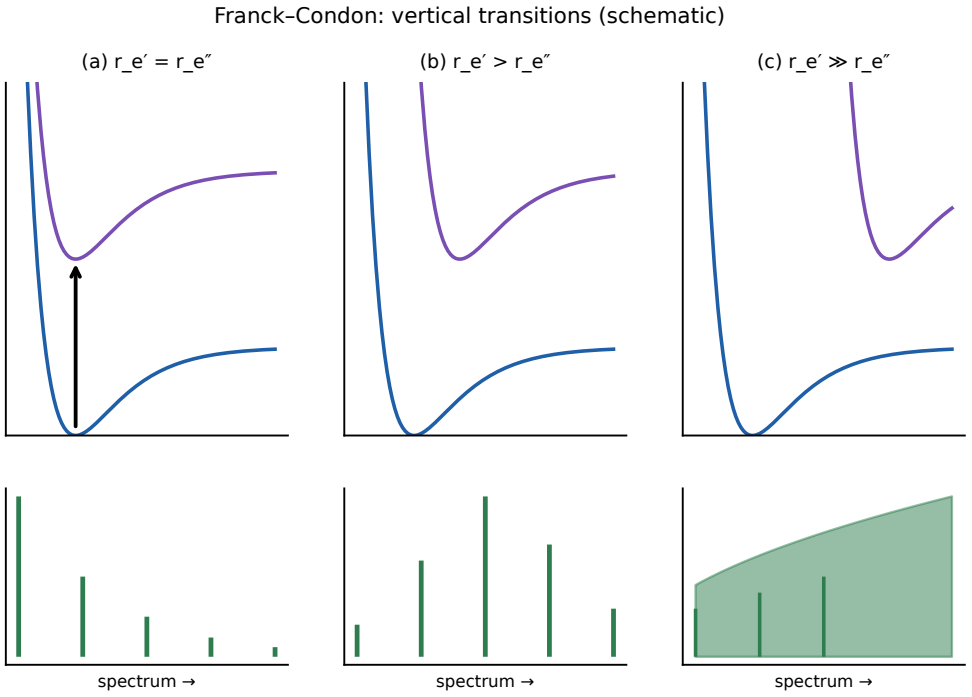


Fig. G21 — Franck–Condon, three cases (a, b, c) with spectra.

In an exam: Draw vertical arrow from v″=0; (c) ends in a continuum.

M2 · Phase Equilibria

T8 · Basic terms & phase rule

No graph for this topic (see formulas/notes).

T9 · Derivation of phase rule

No graph for this topic (see formulas/notes).

T10 · One-component system: water

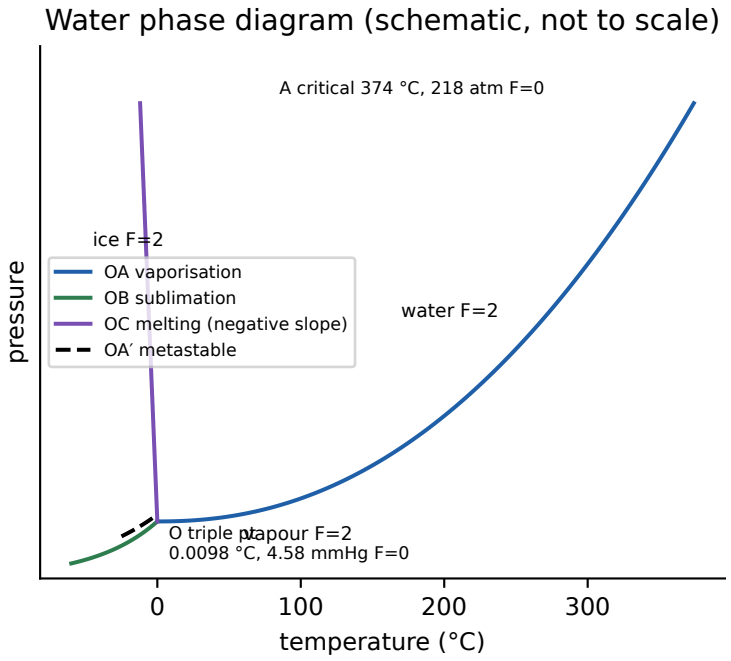


Fig. G22 — Water phase diagram (schematic, not to scale).

In an exam: Label OA, OB, OC, OA′, O, A, with T,P values and F in each region.

T11 · Two-component: reduced phase rule & Pb–Ag

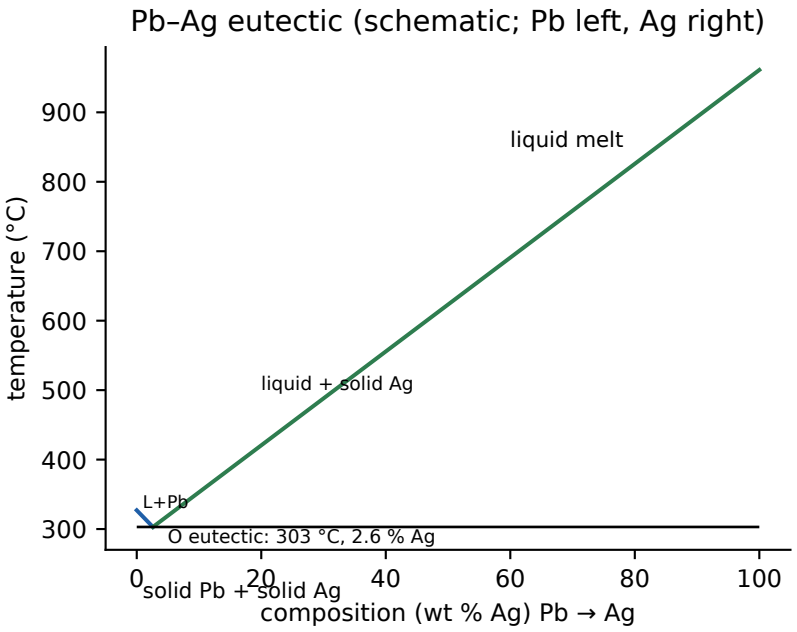


Fig. G23 — Pb–Ag eutectic diagram (schematic; Pb left, Ag right).

In an exam: Eutectic O at 303 °C, 2.6 % Ag; label the four areas. Slide lettering is inconsistent (see conflicts) – follow your faculty's.

T12 · Thermal analysis / cooling curves

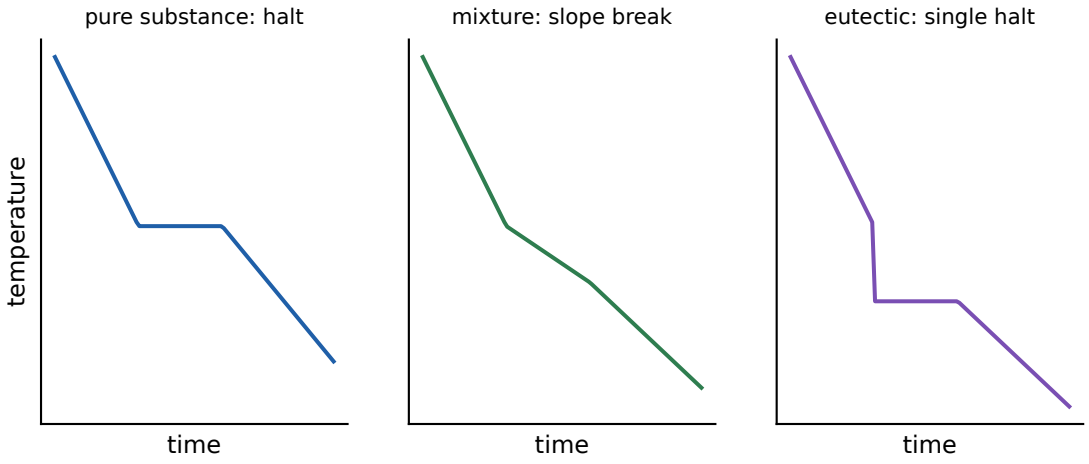


Fig. G24 — Cooling curves: pure, mixture, eutectic.

In an exam: Mark break and halt.

M3 · Computational Chemistry

T13 · Computational chemistry

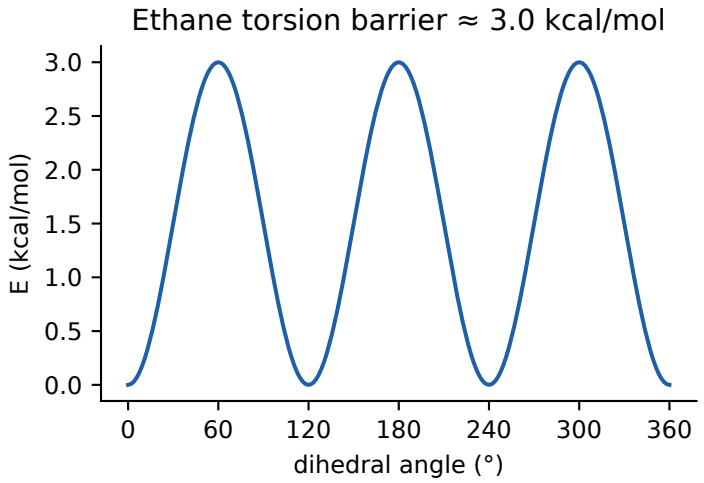


Fig. G25 — Ethane torsion barrier ≈3.0 kcal/mol (illustrative).

In an exam: Staggered minima, eclipsed maxima.